OFFICIAL COLLEGE PROJECT ABSTRACT · PROJECT 32 OF 55

PREDICTIVE MAINTENANCE OF INDUSTRIAL MACHINES USING MULTIVARIATE SENSOR DATA AND DEEP LEARNING

Domain: Industrial IoT & Mechanical Prognostics | **Module:** project_32_predictive_maintenance

OFFICIAL ABSTRACT

Predictive Maintenance of Industrial Machines Using Multivariate Sensor Data and Deep Learning is a comprehensive and crucial project aimed at addressing unexpected equipment failures, costly downtime, and inefficiencies in modern manufacturing and industrial environments. As industrial automation accelerates, facilities increasingly rely on interconnected machinery that continuously generates complex operational metrics, making them prime targets for sudden mechanical breakdowns if faults are not proactively identified. Traditional maintenance methods often rely on scheduled or reactive servicing, which can be time-consuming, resource-intensive, and fundamentally inefficient against rapidly developing, complex mechanical anomalies and subtle component wear. To address these persistent challenges and limitations, this study proposes an advanced, multi-layered approach combining cutting-edge deep learning techniques with intelligent multivariate analysis algorithms for efficient, automated, and real-time machine health monitoring and predictive maintenance.The system utilizes specialized deep learning architectures to continuously capture and extract key behavioral features from extensive multivariate sensor streams, such as fluctuating temperature readings, vibration patterns, pressure variations, and operational speed deviations. Once these underlying structural patterns are mapped, the extracted features are fed into classification and forecasting algorithms designed to enhance decision-making, optimize computational overhead, improve prediction accuracy, and drastically reduce false alarm and misclassification rates. By profiling what constitutes normal, routine operation for each distinct industrial machine, the model establishes a dynamic baseline that adapts to changing workload requirements over time.

📌 Project Summary & System Specifications

Project Title	PREDICTIVE MAINTENANCE OF INDUSTRIAL MACHINES USING MULTIVARIATE SENSOR DATA AND DEEP LEARNING
Specialization / Category	Industrial IoT & Mechanical Prognostics
Target Technology Stack	Python (Flask), PyTorch / TensorFlow / Scikit-Learn, Modern Web UI
Default Backend Port	`:5032`
Local Implementation	`project_32_predictive_maintenance`